DO NOW — RECALL

Thinking procedurally: ordering subprocedures

Name: _____ Class: _____ Date: _____

1. Why does `age>12 && age<18` reject two valid ages for a range 12–18 inclusive?

2. What is a subprocedure?

3. What is an object's state?

4. Why not send a success message before saving a transaction?

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

Because strict comparisons exclude both 12 and 18.

2. Model answer

A named, reusable piece of a larger task.

3. Model answer

The stored values of its properties or fields.

4. Model answer

The save might fail even though the user has already been told it succeeded.

Vocabulary — know these terms

decomposition — Split a problem into smaller operations.

subprocedure — Named step used within a larger solution.

dependency — Stage that must precede another.

state update — Change the stored record or value.

transaction — Coordinated operations that must succeed reliably.

NEW LEARNING

Read, annotate and explain

Decompose the task

Separate the problem into input validation, checking availability, updating the record and notifying the user.

Identify dependencies

No booking should be saved until seats are available; no confirmation should be sent before a successful save.

Make smaller operations reusable

ValidateRequest and SendConfirmation can become independently testable subprocedures.

Explain one step in your own words, using the correct technical terms.

I DO — WORKED EXAMPLE

Follow the worked method

Put password-reset stages in a safe order.

Predict first

First identify the account and create a single-use reset token.

Model the reasoning

Check the token before accepting a new password.

Model the reasoning

Save the change successfully before confirming to the user.

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

Order these steps for ticketing: record purchase, take request, check payment, issue ticket.

Question 2

Design four named subprocedures for booking a classroom and give two key ordering requirements.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

A booking code subtracts seats from a local int parameter but never updates the central system. Identify three problems and three remedies.

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

A booking confirmation can safely be sent before the record is saved.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.